

LLM as GNN: Graph Vocabulary Learning for Text-Attributed Graph Foundation Models

Xi Zhu^{1*}, Haochen Xue^{2*}, Ziwei Zhao³, Wujiang Xu¹, Jingyuan Huang¹,
Minghao Guo¹, Qifan Wang⁴, Kaixiong Zhou⁵, Yongfeng Zhang^{1†}

¹ Rutgers University ² University of Liverpool

³ University of Science and Technology of China

⁴ Meta AI ⁵ North Carolina State University

Abstract

Text-Attributed Graphs (TAGs), where each node is associated with text descriptions, are ubiquitous in real-world scenarios. They typically exhibit distinctive structure and domain-specific knowledge, motivating the development of a Graph Foundation Model (GFM) that generalizes across diverse graphs and tasks. Despite large efforts to integrate Large Language Models (LLMs) and Graph Neural Networks (GNNs) for TAGs, existing approaches suffer from decoupled architectures with two-stage alignment, limiting their synergistic potential. Even worse, existing methods assign out-of-vocabulary (OOV) tokens to graph nodes, leading to graph-specific semantics, token explosion, and incompatibility with task-oriented prompt templates, which hinders cross-graph and cross-task transferability. To address these challenges, we propose PromptGFM, a versatile GFM for TAGs grounded in graph vocabulary learning. PromptGFM comprises two key components: (1) Graph Understanding Module, which explicitly prompts LLMs to replicate the finest GNN workflow within the text space, facilitating seamless GNN-LLM integration and elegant graph-text alignment; (2) Graph Inference Module, which establishes a language-based graph vocabulary ensuring expressiveness, transferability, and scalability, enabling readable instructions for LLM fine-tuning. Extensive experiments demonstrate our superiority and transferability across diverse graphs and tasks. The code is available at this URL ¹.

两个模块：
图理解模块
图推理模块

1 Introduction

Graphs, characterized by their non-Euclidean structures and rich domain-specific knowledge, serve as fundamental representations of complex relational data. Many of them integrate textual information at the node level, forming **Text-Attributed Graphs**

(TAGs), such as citation networks (Eto, 2019; Hu et al., 2020; Buneman et al., 2021), social networks (Kempe et al., 2003; Myers et al., 2014), and molecular graphs (Wieder et al., 2020; Jin et al., 2024a). However, existing solutions heavily depend on task- or dataset-specific training and deployment, limiting their transferability. To address this, we aim to build a **Graph Foundation Model (GFM)** capable of generalizing across different graphs and tasks for TAGs (Mao et al., 2024; Xia et al., 2024).

目前的模型
都是基于特定任务或数据集，无法泛化

In TAGs, Graph Neural Networks (GNNs) and Large Language Models (LLMs) are employed to handle different modalities, as illustrated in Figure 1. (a) **GNN for LLM**. GNNs generate structure-aware node embeddings that complement original textual embeddings, improving LLM inference (Tang et al., 2024; Chai et al., 2023; Liu et al., 2024b). (b) **LLM for GNN**. LLMs extract additional semantic features or labels from textual data, serving as supervision signals for GNN training (Chen et al., 2024c; Liu et al., 2024a; Zhu et al., 2024). However, current loosely coupled architectures with two-stage alignment struggle to fully exploit the synergy between GNNs and LLMs, resulting in suboptimal graph-text alignment in TAGs.

Recently, a noteworthy trend has emerged toward implementing **LLM as GNN**, where graph verbalizers convert graph data into code-like or

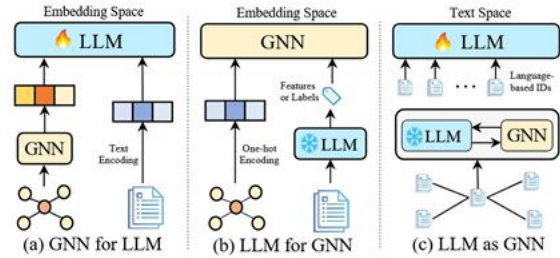


Figure 1: Overview of three GNN-LLM integration paradigms for graph-text alignment: (a) **GNN for LLM** and (b) **LLM for GNN** use decoupled architectures in the embedding space, while (c) our work functions **LLM as GNN** in the text space.

*Equal contribution.

†Corresponding email: yongfeng.zhang@rutgers.edu.

¹<https://github.com/agiresearch/PromptGFM>

heuristic prompts, enabling LLMs prompts to understand graph semantics and structure (Ye et al., 2024; Wang et al., 2024a; Chen et al., 2024a). However, we argue that currently there are no true examples of this category, as they lack the essence of a genuine GNN: *the message-passing paradigm*. As shown in Figure 2(a), a traditional GNN layer includes neighbor sampling, aggregation-update, and optimization (Kipf and Welling, 2017; Velickovic et al., 2017). By stacking multiple layers, structure-less embeddings are gradually transformed into structure-rich embeddings with higher-order signals. Due to the absence of these key components, an urgent challenge arises: *Can we leverage LLMs to faithfully replicate GNNs to capture both graph semantics and structures simultaneously?*

Meanwhile, existing works intuitively treat each graph node as an out-of-vocabulary (OOV) token, leading to graph-specific semantics and uncontrolled token explosion (Tang et al., 2024; Ye et al., 2024). Worse still, due to vocabulary mismatches, ID-based node embeddings within graphs and language-based token embeddings from templates reside in different feature spaces, resulting in semantic misalignment in LLM inference. This incompatibility further hinders the transferability and scalability of graph-specific knowledge across other graphs and tasks. To enable knowledge transfer, a critical challenge emerges: *Can we replace OOV tokens with compatible and universal node representations to build a versatile GFM?*

A versatile GFM should be grounded in **graph vocabulary learning** (Mao et al., 2024; Cai, 2024), ensuring the following properties: (1) **Expressiveness**: it encapsulates both semantic and structural information across diverse graphs. (2) **Transferability**: Every node in any graph should be representable using one or more fundamental units within the vocabulary. (3) **Scalability**: the vocabulary should be inclusive to accommodate unseen nodes, even those beyond existing graphs. Since natural language is a highly expressive medium made up of meaningful and transferable tokens (Raffel et al., 2020; Radford et al., 2021; Palo et al., 2023; Wang et al., 2024c), we are inspired to establish a universal graph vocabulary within the text space for node representations. Thus, we propose PromptGFM as a GFM for TAGs as follows.

Graph Understanding Module. To function LLMs as GNNs, we initialize node features with textual attributes and prompt LLMs to explicitly replicate the fine-grained GNN workflow within

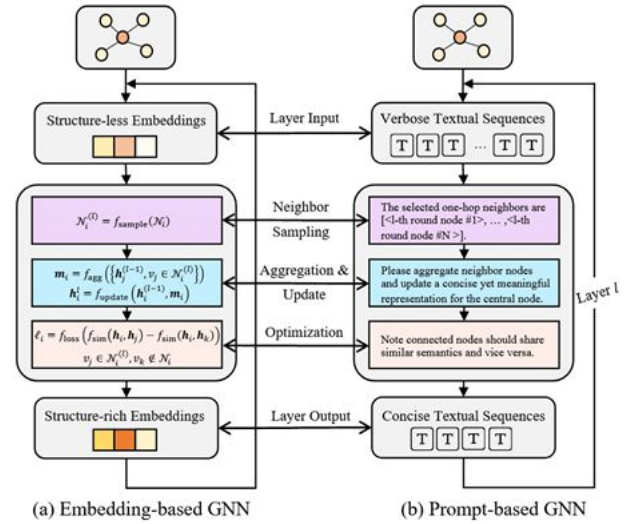


Figure 2: LLM-driven replication of the GNN workflow. We achieve fine-grained alignment between traditional embedding-based GNN and our prompt-based GNN.

the text space. As illustrated in Figure 2, we represent graph structure through one-hop neighbor descriptions and design prompts to guide a flexible aggregation-update mechanism. To optimize, we incorporate heuristic prompts to reflect contrastive loss in each layer. Finally, *iterative LLM calls simulate message passing, progressively refining verbose textual features into concise yet meaningful textual representations rather than numerical embeddings*. Aligning with embedding-based GNN, our prompt-based GNN successfully preserves node semantics while capturing higher-order connections. Consequently, LLMs can operate as GNNs, and GNNs can be interpreted as LLMs, unlocking the full potential of GNN-LLM integration and empowering elegant graph-text alignment.

Graph Inference Module. Having captured semantic and structural information through prompt-based GNNs, we decouple textual representations to establish a graph vocabulary, where each node is mapped to a finite sequence of language-based tokens, essentially as language-based IDs. This vocabulary is universally transferable and scalable across graphs, which resolves semantic irrelevance and graph-text incompatibility. Afterward, these language-based IDs can be incorporated to generate readable and coherent pure-language prompts. Within a multi-instruction fine-tuning framework, we collect diverse instructions across graphs and tasks to effectively fine-tune an LLM, enabling cross-graph and cross-task knowledge transfer. In conclusion, this graph vocabulary eliminates incompatibility and paves the way for general GFMs.

迭代LLM调用模拟消息传递，逐步将冗长的文本特征细化为简洁而有意义的文本表示，而不是数字嵌入。

使用prompt-based GNNs用文本信。

该词汇表在图中具有普遍的可转移性和可扩展性，解决了语义无关性和图文本不兼容的问题。

目前的LLM as GNN模型没有类似于GNN中的信息传递

能否利用LLM复制GNN，同时捕获图的语义和结构

1.图特定语义问题 2.token爆炸 3.不同的特征空间

能否用兼容和通用的节点表示来替换OOV令牌

2 Related Works

GNN-LLM Integration. LLMs have unlocked unprecedented potential for graph machine learning, driving efforts to integrate GNNs and LLMs for modeling TAGs. **(a) GNN for LLM.** As structure tokenizers, GNNs and graph transformers embed graph topology into textual representations, enhancing semantical understanding in LLMs (Tang et al., 2024; Chai et al., 2023; Liu et al., 2024b). However, effectively coordinating GNN-LLM architectures and co-training remains a challenge. **(b) LLM for GNN.** LLMs assist GNNs by generating node-level and edge-level labels to address data sparsity issues (Chen et al., 2024c; Guo et al., 2024; Xia et al., 2024; Shu et al., 2024). Meanwhile, the other line directs LLMs to produce additional features or explanations to overcome semantic deficiencies (Liu et al., 2024a; Zhu et al., 2024; He et al., 2024). However, their reliance on LLM-generated content inevitably introduces noise, impacting performance. **(c) LLM as GNN.** This paradigm directly operates LLMs as GNNs by designing structure verbalizers to convert graph data into code-like or heuristic prompts for LLM inference (Chen et al., 2024a; Ye et al., 2024; Wang et al., 2024a,b). Yet, they fail to capture high-order connections due to the lack of an intrinsic GNN mechanism. Overall, the current decoupled approaches rely on two-stage alignment, failing to fully exploit the strengths of both models. This limitation motivates us to propose a new paradigm where LLMs inherently function as GNNs, maximizing their synergistic potential.

Graph-Text Alignment in Embedding Space.

Current approaches to graph-text alignment primarily operate in the embedding space. One approach uses graph encoders as prefixes, mapping graph-aware embeddings to language-based embeddings for LLMs fine-tuning in a shared space (Huang et al., 2023, 2024). Other works adopt a two-tower architecture, leveraging contrastive learning (Li et al., 2023a; Brannon et al., 2023; Tang et al., 2024; Jin et al., 2024b), iterative training (Zhao et al., 2023a; Zhu et al., 2024), or knowledge distillation (Mavromatis et al., 2023) to align distinct representations. However, these methods encounter a persistent modality gap, limiting transferability and scalability across graphs. Since TAGs inherently contain textual information, we advocate for shifting graph-text alignment to the text space.

Graphs Foundation Models. A GFM aims to achieve transferability across different datasets and

GFM旨在实现不同数据集和任务之间的可转移性

tasks, where the key challenge lies in finding a graph vocabulary that identifies transferable units to encode invariance on graphs (Liu et al., 2023; Mao et al., 2024). Previous works rely on domain-specific vocabularies: GraphGPT (Tang et al., 2024) assumes unique IDs to nodes and creates a dataset-specific vocabulary, while MoleBERT (Xia et al., 2023) defines a molecular graph vocabulary by converting atomic properties into chemically meaningful codes. Despite their success, they lack in-context learning and cross-domain transferability. Recently, while some studies have investigated understanding and inferring graphs in natural language (Fatemi et al., 2024; Wang et al., 2023; Zhao et al., 2023b; Liu et al., 2024a; Zhang et al., 2024b,a), none of them have attempted to establish a language-based graph vocabulary. To address this, we introduce an expressive graph vocabulary to exploit the inherent transferability of natural language, advancing the development of a versatile GFM.

关键在于找到一个图词汇表来识别可转移单元,以编码图上的不变性

3 Preliminaries

V:点集 E:边集 X:节点文本属性

Text-Attributed Graphs. A TAG is formally represented as $G = (V, E, X)$, where V is the set of nodes, E is the set of edges. In this work, each node $v_i \in V$ is associated with a textual description $X_i = (x_i^1, x_i^2, \dots, x_i^{n_i})$, where each $x_i^k \in \mathcal{X}, k = 1, \dots, n_i$. Here, X denotes the textual attributes for nodes, and $\mathcal{X}$ represents the natural language token dictionary. X:自然语言标记字典

Graph Neural Networks. GNNs have emerged as state-of-the-art models in graph machine learning, predominately relying on the message-passing paradigm. In practice, a GNN first selects neighboring nodes to a target node, aggregates their representations to capture local structure, and then updates the target node’s representation. Mathematically, for a given node v_i , the l -th layer of a general GNN can be formulated as:

$$\begin{aligned} \mathcal{N}_i^{(l)} &= f_{\text{sample}}(\mathcal{N}_i), \quad \text{邻居采样} \\ \mathbf{m}_i^{(l)} &= f_{\text{agg}}\left(\left\{\mathbf{h}_j^{(l-1)}, v_j \in \mathcal{N}_i^{(l)}\right\}\right), \quad \text{聚合} \\ \mathbf{h}_i^{(l)} &= f_{\text{update}}\left(\mathbf{h}_i^{(l-1)}, \mathbf{m}_i^{(l)}\right) \quad \text{更新} \end{aligned} \quad (1)$$

where $\mathbf{h}_i^{(l)}$ is the node embedding of v_i in the l -th layer. $\mathcal{N}_i$ denotes full neighbors and $\mathcal{N}_i^{(l)}$ is sampled neighbors in the l -th layer. To capture high-order connections, we stack L layers and derive final embeddings as $\mathbf{h}_i = f_{\text{pooling}}\left(\mathbf{h}_i^{(1)}, \dots, \mathbf{h}_i^{(L)}\right)$

Ni:邻居

prefix-tuning

two-tower

模态差异限制了跨图的可转移性和可扩展性

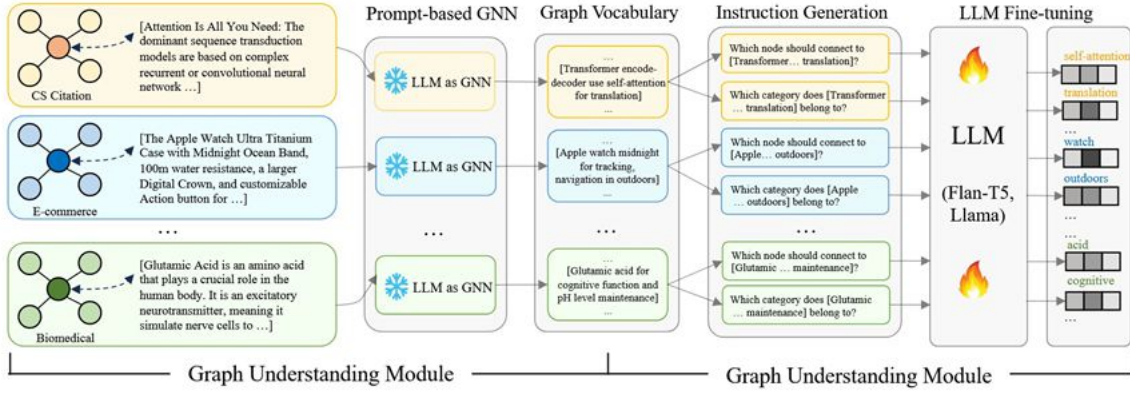


Figure 3: The pipeline of PromptGFM. (a) Graph Understanding Module: For arbitrary TAGs from different domains, prompt-based GNN replicates traditional embedding-based GNN workflow in the text space, generating compact node representations. (b) Graph Inference Module: We establish a unified graph vocabulary and extract language-based IDs to generate massive pure-language prompts, enabling LLM fine-tuning across graphs and tasks.

(Grattarola et al., 2024). For optimization, a contrastive loss with negative sampling is commonly used in unsupervised graph learning (Hamilton et al., 2017; Velickovic et al., 2019), expressed as:

$$\ell = f_{\text{loss}}(f_{\text{sim}}(\mathbf{h}_i, \mathbf{h}_j), f_{\text{sim}}(\mathbf{h}_i, \mathbf{h}_k)), \quad (2)$$

where $v_j \in \mathcal{N}_i^{(l)}$ is a positive sample and $v_k \notin \mathcal{N}_i$ is a negative one. $f_{\text{sim}}(\cdot)$ measures the similarity between two nodes. $f_{\text{loss}}(\cdot)$ enforces contrastive learning by increasing similarity for connected nodes and reducing it for unconnected ones.

4 Methodology

This section describes the pipeline of our proposed PromptGFM, as illustrated in Figure 3.

4.1 Graph Understanding Module

The graph understanding module aims to generate expressive representations for each node within the graph, supporting the subsequent inference module. The main challenge lies in effectively aligning the semantic and structural information, where LLMs excel in textual understanding and GNNs in structural modeling (Li et al., 2023b; Ren et al., 2024). To bridge the gap in GNN-LLM integration, we propose prompt-based GNNs that operate LLMs as GNNs by prompting LLMs to replicate general GNN workflow within the text space.

GNN Replication with LLMs. Our priority is to design appropriate prompts that enable LLMs to function as GNNs. This requires considering three essential factors: (a) **Graph Representations:** How can we effectively describe the node semantics and graph structure to LLMs? (b) **Graph Structure:** How can we incorporate message passing to encode structural dependencies? (c) **Graph Semantics:** How can we distill core semantics into concise yet expressive textual representations?

As illustrated in Figure 2, we perform a fine-grained replication of GNN, i.e. prompt-based GNN. First, we initialize node representations by summarizing textual attributes, akin to look-up layers in traditional GNNs. Similarly, we sample its one-hop neighbor information to reduce computational overhead as follows:

$$\{X_j^{(l-1)}, \{v_j\} \subset \mathcal{N}_i\} \leftarrow \text{Prompt}_{\text{sample}}(X^{(l-1)}, \mathcal{N}_i), \quad (3)$$

where $X^{(l-1)} = \{X_0^{(l-1)}, X_1^{(l-1)}, \dots, X_{|V|-1}^{(l-1)}\}$ denotes textual representations of all nodes in the previous layer, and $\{X_j^{(l-1)}\}$ corresponds to the selected neighbors in $\mathcal{N}_i$. In practice, we limit neighbor sampling to a maximum of 20 to mitigate prompt length constraints. With this context, we directly prompt LLMs to perform essential message aggregation-update process, formulated as:

$$X_i^{(l)} \leftarrow \text{Prompt}_{\text{agg-upd}}(\{X_j^{(l-1)}, \{v_j\} \in \mathcal{N}_i^{(l)}\}, X_i^{(l-1)}), \quad (4)$$

where we seek flexible message passing mechanism without specification, moving beyond traditional operators like mean and weighted aggregators. In terms of optimization, we adopt an unsupervised graph learning with contrastive loss to accommodate various downstream tasks, increasing neighbor similarity while separating distant nodes. Since negative sampling is redundant in the situation, we use LLM prompts to intuitively steer this process. Prompts can be found in Appendix D.

After repeating L rounds, we obtain the final textual representation as $T_i = X_i^{(L)}$. These representations encapsulate both semantic and structural information, effectively solving the outlined issues: (a) **Graph Representations:** One-hop neighbor descriptions are equivalent to an adjacency matrix, representing the entire graph structure. (b)

floss通过增加连接节点的相似性并减少非连接节点的相似性来加强对比学习

prompt-based GNN通过提示LLM在文本空间内复制GNN的操作

图表示：如何有效地向LLM描述节点语义和图结构 图结构：一跳邻居描述等价于邻接矩阵，表示整个图结构 图语义：使合消息传递来编码结构依赖关系 图语义：如何将核心语义提炼成简短相同的提示迭代调用LLM 图语义：明确指示LLM为每个节点生成简洁而富有表现力的文本表示

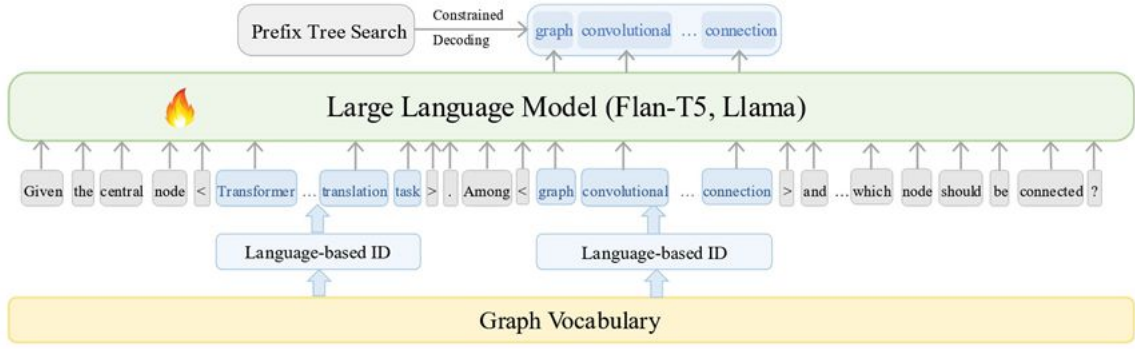


Figure 4: An instance of graph inference module in link prediction, where language-based IDs are indexed from the graph vocabulary to generate readable instructions using task-oriented templates. We adopt multi-instruction fine-tuning framework to unify diverse graphs and tasks and learn transferable knowledge for GFM.

Graph Structure: To capture high-order relationships, we iteratively invoke LLMs using the same prompts, feeding each round’s output into the next.

(c) Graph Semantics: Simultaneously, we explicitly instruct LLMs to produce concise yet expressive textual representations for each node, gradually refining them for denser and richer semantics.

We systematically examine how prompt-based GNNs faithfully mirror embedding-based GNNs in Appendix C. In summary, our fine-grained replication exemplifies the potential of using LLMs as GNNs, fostering seamless GNN-LLM integration and elegant graph-text alignment.

4.2 Graph Inference Module

The graph inference module seeks to unify diverse graphs and tasks and acquire transferable knowledge through multi-instruction fine-tuning using LLMs. Due to distinct vocabularies, existing methods, which represent nodes as OOV tokens, suffer from semantic misalignment between ID-based and language-based embeddings in task-oriented prompts, constraining their transferability. To tackle this limitation, we introduce a novel language-based graph vocabulary that bridges this incompatibility, enabling massive readable and coherent instructions to support LLM inference.

Graph Vocabulary Learning. Learning a transferable graph vocabulary, whose fundamental units can represent each node, is key to building GFMs. Its effectiveness depends on three essential criteria: expressiveness, transferability, and scalability. Since each node has been associated with a textual representation that encapsulates its core semantics and local structure, we intuitively propose a universal language-based graph vocabulary using these rich representations. In this vocabulary, each node is represented by a finite sequence of language tokens, i.e. a language-based ID, defined as follows:

$$\mathcal{F} : V \rightarrow T, \quad (5)$$

where each graph node v_i is assigned a sequence as $T_i = (t_i^1, t_i^2, \dots, t_i^{m_i})$, where $t_i^k \in \mathcal{X}$ and $\mathcal{X}$ is a dictionary of general natural language tokens. Along this line, all nodes in V have language-based IDs as $T = \{T_0, T_1, \dots, T_{|V|-1}\}$. Consequently, natural language tokens form the fundamental building blocks of our graph vocabulary, with structured token sequences representing graph nodes, analogous to words in standard lexicons.

Our graph vocabulary meets all expected criteria: **(a) Expressiveness.** Language-based IDs preserve rich semantic and structural information from an open-world setting. **(b) Transferability.** Like human lexicons, it shares natural language as a common foundation, ensuring inherent cross-graph transferability. **(c) Scalability.** Any node, whether previously seen or not, can dynamically generate its language-based ID, ensuring compatibility with existing nodes and mitigating token explosion.

Instruction Fine-Tuning. We employ a multi-instruction fine-tuning framework to incorporate various graphs and tasks (Chung et al., 2024; Wei et al., 2022). As illustrated in Figure 4, we index nodes from the graph vocabulary and embed their language-based IDs into task-oriented prompt templates to construct completed instructions $\mathcal{T}$:

$$\mathcal{T} \leftarrow \text{Prompt}_{\text{template}}(T, G | \mathcal{F}), \quad (6)$$

which include language-based IDs of the central nodes and their respective local structure T from G , depending on specific graph-centric tasks (e.g., node classification or link prediction). Thus, these instructions are fully readable and coherent, as they are composed entirely of natural language tokens. Afterward, we convert all the question-answering tasks into a unified text-to-text format for LLM fine-tuning (Mishra et al., 2022). Given target sequences Y , the loss function is computed as:

$$\mathcal{L} = - \sum_{j=1}^{|Y|} \log \Pr(Y_j | \mathcal{T}, Y_{<j}), \quad (7)$$

$\Pr(Y_j | \mathcal{T},$

$Y_{<j})$: 输出序列 Y 中第 j 个 Y_j 的token概率, 条件是指令 $\mathcal{T}$ 和所有先前

每个节点 v_i 的结构感知语义表示 T_i , 每个 tik 是一个自然语言token, 可以是词、短语、子词 (取决于tokenizer)

表达力 迁移性 可扩展性

图推理模块旨在统一不同的图和任务, 并通过使用LLM的多指令微调来获取可转移的知识

每个节点都由有限的语言标记序列表示, 即基于语言的ID

where $\Pr(Y_j | \mathcal{T}, Y_{<j})$ is the probability of the j -th token Y_j in the output sequence Y , conditioned on the instruction $\mathcal{T}$ and all previous tokens $Y_{<j} = (Y_1, Y_2, \dots, Y_{j-1})$. Using T5 (Raffel et al., 2020), FLAN (Wei et al., 2022), or Llama (Touvron et al., 2023) as the backbone, we can co-train a unified LLM across diverse graphs and tasks by fine-tuning, which enables the acquisition of open-world global knowledge and inclusive accommodation of unseen graphs or tasks.

Constrained Decoding via Prefix Tree Search.

To mitigate LLM hallucination in link prediction, we introduce a constrained decoding method using a prefix tree search strategy to regulate LLM outputs (Cao et al., 2021; Tan et al., 2024). Specifically, we craft a prefix tree from language-based IDs of all candidate nodes, where each tree node represents a natural language token. Each unique path from the root to a leaf corresponds to a specific language-based ID. During autoregressive generation, token output is restricted to a valid path within the prefix tree, ensuring predictions align with actual graph nodes and eliminating hallucinations. This effectiveness is attributed to the discrete nature of language-based IDs, further highlighting the flexibility of the proposed graph vocabulary.

5 Experiments

In this section, we conduct extensive experiments to address the following research questions (RQs):

- **RQ1:** How does PromptGFM perform on supervised node classification and link prediction?
- **RQ2:** How is the transferability across diverse graphs and tasks as a versatile GFM?
- **RQ3:** How does each module contribute to overall performance of PromptGFM?
- **RQ4:** What affect the GNN replication?

Datasets. We present the following datasets covering three domains: (a) computer science: Cora (McCallum et al., 2000), Citeseer (Giles et al., 1998), Ogbn-arxiv (Hu et al., 2020), and WikiCS (Mernyei and Cangea, 2020); (b) e-commerce: Photo and History (Ni et al., 2019; Yan et al., 2023); (c) biomedical: PubMed (Sen et al., 2008). Details can be found in Appendix A.

Baselines. We make comprehensive comparisons with existing methods in four categories: (1) **Graph-agnostic methods.** We consider the MLP model without graph structure. (2) **GNN-based methods.** We employ four fundamental GNN models: GCN (Kipf and Welling, 2017), GAT

(Velickovic et al., 2017), GraphSAGE (Hamilton et al., 2017), and ReVGNN (Li et al., 2021). We also explore SGFormer (Wu et al., 2023) and NodeFormer (Wu et al., 2022), which leverage transformer architectures to model graph data. (3) **BERT-based methods.** We utilize BERT (Devlin et al., 2019) and Sentence-BERT (Reimers and Gurevych, 2019) to obtain textual representations for downstream tasks. (4) **GNN-LLM Integration methods.** Following the proposed taxonomy, we select GraphPrompter (Liu et al., 2024b) as an instance of using GNNs to enhance LLMs. OFA (Liu et al., 2024a) and ENGINE (Zhu et al., 2024) are examples of leveraging LLMs for GNNs. Besides, we incorporate LLaGA (Chen et al., 2024a) as an attempt of implementing LLM as GNN. We provide baseline details in Appendix B.

Reproduction Settings. We implement PromptGFM in PyTorch and run all experiments on four NVIDIA RTX A6000 GPUs. The graph understanding module leverages OpenAI’s GPT-4o mini while we fine-tune a Flan-T5 or Llama3-8B in the graph inference module. For evaluation metrics, we use accuracy and Macro-F1 (abbreviated as M-F1) for node classification, and accuracy and HR@1 for link prediction, respectively. We provide further implementation details in Appendix E.

5.1 Performance Comparison (RQ1)

Main Results. We train each model independently from scratch on a single graph and compare their performance. Tables 1 and 2 present the results for node classification and link prediction accuracy, respectively. PromptGFM achieves substantial improvements over state-of-the-art models, leading to the following insights. First, graph-based models outperform graph-agnostic ones, highlighting the importance of structural information. Second, our method surpasses OFA, ENGINE, and GraphPrompter in node classification, partly due to their decoupled architectures in both GNN-for-LLM and LLM-for-GNN paradigms. Third, PromptGFM outperforms LLaGA, which relies on templates to understand graph semantics and structures. The results suggest that heuristic prompts alone fail to capture sufficient high-order signals without an actual GNN mechanism. In contrast, PromptGFM demonstrates the potential of leveraging LLMs as GNNs through a prompt-based GNN, establishing a novel paradigm for graph-text alignment in TAGs. **Generative Link Prediction.** Following a transductive setting, we partition the graph by links, cre-

通过前缀
树约束输出

RQ1: PromptGFM如何对监督节点分类和链路预测进行操作? RQ2: 作为一个通用的GFM, 在不同的图形和任务之间的可转移性如何 RQ3: 每个模块对PromptGFM的整体性能有何贡献? RQ4: 什么会影响GNN复制

Table 1: Evaluation results (%) on node classification accuracy ($\uparrow$) for all datasets. We highlight the methods with the **first** and **second** best performances.

Method	Cora	Citeseer	PubMed	Ogbn-arxiv	History	Photo	WikiCS
MLP	62.29	64.42	62.88	62.07	64.62	61.21	68.41
GCN	82.47	76.11	77.36	66.15	81.93	78.58	76.33
GAT	82.92	77.30	74.36	65.29	82.85	82.38	78.21
SAGE	83.69	73.17	83.22	68.78	82.12	80.06	79.56
RevGNN	86.90	77.34	82.16	70.43	83.04	83.24	81.22
SGFormer	82.36	73.76	78.92	63.44	78.98	80.12	76.56
NodeFormer	81.55	72.98	76.49	73.21	79.60	78.51	75.47
BERT	79.02	72.83	76.74	71.90	72.97	68.82	77.98
Sentence-BERT	78.09	72.12	75.48	77.24	74.10	69.02	77.72
OFA	75.72	71.58	75.26	74.68	81.43	84.46	78.02
LLaGA	81.25	68.80	86.54	76.05	82.55	85.34	80.74
ENGINE	91.48	78.46	90.24	76.02	82.46	83.75	81.56
GraphPrompter	80.26	73.61	94.80	75.61	79.42	80.04	80.98
PromptGFM (Flan-T5)	91.72	84.49	92.83	80.58	82.33	85.41	81.49
PromptGFM (Llama3)	92.42	85.32	94.65	83.78	86.72	86.61	84.66

Table 2: Evaluation results (%) of accuracy ($\uparrow$) for link prediction in the discriminative setting.

Method	Discriminative Setting			
	Cora	Citeseer	Obgn-arxiv	PubMed
GCN	77.15	78.72	80.89	77.36
GAT	70.44	77.17	76.25	74.36
SAGE	85.31	87.15	80.76	83.22
GraphPrompter	90.10	91.67	73.21	80.49
Ours (Flan-T5)	90.57	92.03	81.12	87.64
Ours (Llama3)	91.68	93.46	84.27	88.12

ate an input graph from the training set, and finally predict unseen connections for test nodes, with existing nodes as potential candidates. From Table 3, PromptGFM consistently outperforms traditional GNN models. Unfortunately, existing GNN-LLM approaches overlook this setting due to their reliance on OOV token embeddings, preventing LLM outputs from mapping to specific nodes and leading to unresolved hallucination issues. In contrast, our framework represents nodes as finite token sequences, thereby allowing constrained decoding to regulate LLM outputs. This further underscores the critical role of our graph vocabulary and the flexibility and scalability of PromptGFM.

5.2 Cross-graph Transferability (RQ2)

We examine cross-graph transferability in both intra-domain and inter-domain scenarios. Following (Chen et al., 2024b), we explore two settings: (1) *Pre-training*, where models transfer to unseen datasets without prior exposure during training, and (2) *Co-training*, where available target data is also integrated for joint training with auxiliary datasets. This results in four distinct transfer paradigms.

Intra-domain Cross-graph Transferability. As shown in Table 4, all settings surpass the off-the-

Table 3: Evaluation results (%) of HR@1 ($\uparrow$) for link prediction in the generative setting.

Method	Generative Setting		
	Cora	Citeseer	PubMed
GCN	5.95	6.82	0.51
GAT	2.22	3.59	0.28
SAGE	6.59	8.73	0.45
PromptGFM	8.21	8.90	1.21

shelf LLM. Meanwhile, incorporating Arxiv yields greater gains than Cora, suggesting that larger and richer datasets enhance transferability more effectively. In the *pre-training* setting, significant improvements over direct inference with an off-the-shelf LLM highlights strong zero-shot transferability within the computer science domain, while similar results on *co-training* further confirm effective knowledge transfer. These findings underscore the potential to collect more graph data and build more comprehensive and knowledgeable GFMs.

Inter-domain Cross-graph Transferability. We outline cross-domain results in Table 6. In the *pre-training* setting, while incorporating a single external graph enhances performance, adding multiple graphs can lead to negative transfer. More critically, in the *co-training* setting under supervised learning, performance deteriorates as more cross-domain data is introduced. This decline stems from domain-specific knowledge conflicts, potentially caused by catastrophic forgetting or incompatible hyperparameters that fail to accommodate to varying domain distributions. Although PromptGFM provides a unified interface for diverse graphs and tasks, we leave robust positive knowledge transfer as future work, further discussed in Section 7.

Cross-task Transferability. We demonstrate the transfer performance from link prediction to node

Table 4: Intra-domain cross-graph generalization. We transfer from the computer science domain to the Cite-seer dataset for node classification task. The first row indicates direct inference using off-the-shelf LLM.

Source		Pre-training		Co-training	
Cora	Arxiv	Acc(↑)	M-F1(↑)	Acc(↑)	M-F1(↑)
✗	✗	27.64	17.10	84.49	82.31
✓	✗	51.63	45.10	84.96	83.22
✗	✓	60.34	54.81	85.45	83.91
✓	✓	61.25	55.66	86.77	84.24

Table 5: Cross-task generalization. $LP \rightarrow NC$ means training on link prediction (LP) while testing on node classification (NC). $ST-NC$ refers to supervised training on the NC task from scratch.

Setting	Cora		Citeseer		PubMed	
	Acc(↑)	M-F1(↑)	Acc(↑)	M-F1(↑)	Acc(↑)	M-F1(↑)
<i>zero-shot</i>	18.54	12.16	27.64	17.10	39.12	39.84
$LP \rightarrow NC$	60.74	55.42	50.12	44.68	57.42	58.79
$ST-NC$	91.72	90.06	84.49	80.13	90.67	91.82

classification in Table 5. As expected, $LP \rightarrow NC$ generally surpasses *zero-shot*, showing successful knowledge transfer to unseen tasks. However, its performance is substantially lower than $ST-NC$, indicating that cross-task transfer is more challenging than cross-graph generalization. Overall, these results validate PromptGFM’s ability to effectively transfer knowledge across graphs and tasks, establishing it as a versatile and knowledgeable GFM.

5.3 Ablation Studies (RQ3)

To assess the contribution of each module, we introduce three ablated variants: (1) *w/o understanding*, which removes the prompt-based GNN; (2) *w/o inference*, which eliminates multi-instruction fine-tuning; and (3) *w/o both*, which directly prompts LLMs using raw textual attributes. Figure 5 reports node classification accuracy, where our full model consistently outperforms all variants. First, the drop in *w/o understanding* indicates that omitting GNN replication leads to a loss of semantic and structural information. Second, *w/o inference* shows a significant decline, emphasizing the crucial role of LLM fine-tuning in integrating domain-specific knowledge. Lastly, *w/o both* yields the worst results, underscoring the synergy between the understanding and inference modules.

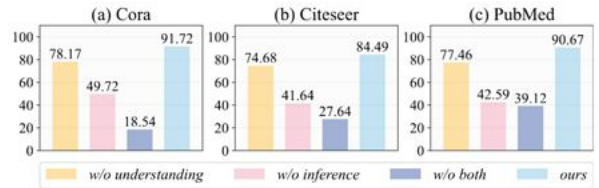
5.4 Exploration Studies (RQ4).

We analyze factors affecting prompt-based GNNs. Since GNNs are permutation-invariant, we evaluate LLM-powered prompt-based GNNs using Flan-T5

Table 6: Inter-domain cross-graph generalization. We transfer from the computer science domain to biomedical domain (i.e. PubMed) on node classification.

Source			Pre-training		Co-training	
Cora	Citeseer	Arxiv	Acc(↑)	M-F1(↑)	Acc(↑)	M-F1(↑)
✗	✗	✗	39.12	39.84	90.67	91.82
✓	✗	✗	51.76	52.84	86.34	87.29
✗	✓	✗	40.12	42.38	85.21	86.12
✗	✗	✓	60.21	62.02	82.17	86.01
✓	✓	✗	50.17	51.74	84.39	81.94
✓	✗	✓	57.28	59.71	81.32	83.14
✗	✓	✓	55.34	57.11	82.13	83.10
✓	✓	✓	53.07	54.90	80.43	80.79

Figure 5: Ablation studies on node classification.



消融实验

Table 7: Permutation sensitivity of prompt-based GNNs.

Variant	Cora	Citeseer
<i>shuffle nodes</i>	90.64 ± 1.12	83.79 ± 0.89
<i>shuffle tokens</i>	90.55 ± 1.03	84.28 ± 1.06

with two variants: (1) *shuffle nodes*, which randomizes neighbor order, and (2) *shuffle tokens*, which injects position-specific tokens to nodes and shuffle them instead. As illustrated in Table 7, regardless of the shuffle method, node classification accuracy exhibits only minor fluctuations, having negligible impact on the stability of prompt-based GNNs.

Additionally, we investigate the impact of layer depth in Appendix F. Furthermore, we present a case study on layer-by-layer representations in Appendix G.1, and compare language-based IDs with keywords from citation datasets in Appendix G.2.

6 Conclusion

In this work, we present PromptGFM, a GFM for TAGs built on graph vocabulary learning. With GNN replication within the text space, we decouple refined textual node representations and establish a unified graph vocabulary. This vocabulary endows compatibility and scalability, facilitating effective LLM fine-tuning with readable instructions for enhanced transferability. Experiments validate its superior performance and effective cross-graph and cross-task generalization. Our research reveal the potential of using LLM as GNN, opening new avenues for developing GFMs in TAGs.

打乱节点 打乱
s

7 Limitations

The proposed PromptGFM framework introduces a promising paradigm for building a Graph Foundation Model (GFM) in Text-Attributed Graphs (TAGs). However, like any novel framework, it has limitations that should be acknowledged for future improvements. First, while we establish a foundation model interface, incorporating a large number of datasets may negatively impact transferability, particularly in cross-domain scenarios. A more stable knowledge transfer mechanism is needed to mitigate domain shifts and preserving generalization. Second, our prompt-based GNN processes each graph independently, disregarding dataset imbalances and cross-graph semantic similarities. One solution is to explore cross-graph semantic alignment and attentive learning for better knowledge integration across heterogeneous graph datasets. Third, our approach remains constrained to text-attributed graphs, leaving a gap in developing a fully text-free graph foundation model. These limitations highlight potential directions for future research in building more efficient and comprehensive GFMs.

References

- William Brannon, Suyash Fulay, Hang Jiang, Won-june Kang, Brandon Roy, Jad Kabbara, and Deb Roy. 2023. [Congrat: Self-supervised contrastive pre-training for joint graph and text embeddings](#). *CoRR*, abs/2305.14321.
- Peter Buneman, Dennis Dosso, Matteo Lissandrini, and Gianmaria Silvello. 2021. [Data citation and the citation graph](#). *Quant. Sci. Stud.*, 2(4):1399–1422.
- Yongqiang Cai. 2024. [Vocabulary for universal approximation: A linguistic perspective of mapping compositions](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net.
- Nicola De Cao, Gautier Izacard, Sebastian Riedel, and Fabio Petroni. 2021. [Autoregressive entity retrieval](#). In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net.
- Ziwei Chai, Tianjie Zhang, Liang Wu, Kaiqiao Han, Xiaohai Hu, Xuanwen Huang, and Yang Yang. 2023. [Graphllm: Boosting graph reasoning ability of large language model](#). *CoRR*, abs/2310.05845.
- Runjin Chen, Tong Zhao, Ajay Kumar Jaiswal, Neil Shah, and Zhangyang Wang. 2024a. [Llaga: Large language and graph assistant](#). In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net.
- Zhikai Chen, Haitao Mao, Jingzhe Liu, Yu Song, Bingheng Li, Wei Jin, Bahare Fatemi, Anton Tsitsulin, Bryan Perozzi, Hui Liu, and Jiliang Tang. 2024b. [Text-space graph foundation models: Comprehensive benchmarks and new insights](#). *CoRR*, abs/2406.10727.
- Zhikai Chen, Haitao Mao, Hongzhi Wen, Haoyu Han, Wei Jin, Haiyang Zhang, Hui Liu, and Jiliang Tang. 2024c. [Label-free node classification on graphs with large language models \(llms\)](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Hyung Won Chung, Le Hou, Shayne Longpre, Barret Zoph, Yi Tay, William Fedus, Yunxuan Li, Xuezhi Wang, Mostafa Dehghani, Siddhartha Brahma, Albert Webson, Shixiang Shane Gu, Zhuyun Dai, Mirac Suzgun, Xinyun Chen, Aakanksha Chowdhery, Alex Castro-Ros, Marie Pellat, Kevin Robinson, Dasha Valter, Sharan Narang, Gaurav Mishra, Adams Yu, Vincent Y. Zhao, Yanping Huang, Andrew M. Dai, Hongkun Yu, Slav Petrov, Ed H. Chi, Jeff Dean, Jacob Devlin, Adam Roberts, Denny Zhou, Quoc V. Le, and Jason Wei. 2024. [Scaling instruction-finetuned language models](#). *J. Mach. Learn. Res.*, 25:70:1–70:53.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. [BERT: pre-training of deep bidirectional transformers for language understanding](#). In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, NAACL-HLT 2019, Minneapolis, MN, USA, June 2-7, 2019, Volume 1 (Long and Short Papers)*, pages 4171–4186. Association for Computational Linguistics.
- Masaki Eto. 2019. [Extended co-citation search: Graph-based document retrieval on a co-citation network containing citation context information](#). *Inf. Process. Manag.*, 56(6).
- Bahare Fatemi, Jonathan Halcrow, and Bryan Perozzi. 2024. [Talk like a graph: Encoding graphs for large language models](#). In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- C. Lee Giles, Kurt D. Bollacker, and Steve Lawrence. 1998. [Citeseer: An automatic citation indexing system](#). In *Proceedings of the 3rd ACM International Conference on Digital Libraries, June 23-26, 1998, Pittsburgh, PA, USA*, pages 89–98. ACM.
- Daniele Grattarola, Daniele Zambon, Filippo Maria Bianchi, and Cesare Alippi. 2024. [Understanding pooling in graph neural networks](#). *IEEE Trans. Neural Networks Learn. Syst.*, 35(2):2708–2718.
- Zirui Guo, Lianghao Xia, Yanhua Yu, Yuling Wang, Zixuan Yang, Wei Wei, Liang Pang, Tat-Seng Chua,

- and Chao Huang. 2024. Graphedit: Large language models for graph structure learning. *CoRR*, abs/2402.15183.
- William L. Hamilton, Zitao Ying, and Jure Leskovec. 2017. Inductive representation learning on large graphs. In *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pages 1024–1034.
- Xiaoxin He, Xavier Bresson, Thomas Laurent, Adam Perold, Yann LeCun, and Bryan Hooi. 2024. Harnessing explanations: Llm-to-llm interpreter for enhanced text-attributed graph representation learning. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Weihua Hu, Matthias Fey, Marinka Zitnik, Yuxiao Dong, Hongyu Ren, Bowen Liu, Michele Catasta, and Jure Leskovec. 2020. Open graph benchmark: Datasets for machine learning on graphs. In *Advances in Neural Information Processing Systems 33: Annual Conference on Neural Information Processing Systems 2020, NeurIPS 2020, December 6-12, 2020, virtual*.
- Xuanwen Huang, Kaiqiao Han, Dezheng Bao, Qianjin Tao, Zhisheng Zhang, Yang Yang, and Qi Zhu. 2023. Prompt-based node feature extractor for few-shot learning on text-attributed graphs. *CoRR*, abs/2309.02848.
- Xuanwen Huang, Kaiqiao Han, Yang Yang, Dezheng Bao, Qianjin Tao, Ziwei Chai, and Qi Zhu. 2024. Can GNN be good adapter for llms? In *Proceedings of the ACM on Web Conference 2024, WWW 2024, Singapore, May 13-17, 2024*, pages 893–904. ACM.
- Mingyu Jin, Haochen Xue, Zhenting Wang, Boming Kang, Ruosong Ye, Kaixiong Zhou, Mengnan Du, and Yongfeng Zhang. 2024a. Prollm: Protein chain-of-thoughts enhanced LLM for protein-protein interaction prediction. *CoRR*, abs/2405.06649.
- Mingyu Jin, Chong Zhang, Liangyao Li, Zihao Zhou, Yongfeng Zhang, et al. 2024b. Attacker: How to evaluate the effectiveness of jailbreak attacking on large language models. *arXiv preprint arXiv:2401.09002*.
- David Kempe, Jon M. Kleinberg, and Éva Tardos. 2003. Maximizing the spread of influence through a social network. In *Proceedings of the Ninth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Washington, DC, USA, August 24-27, 2003*, pages 137–146. ACM.
- Thomas N. Kipf and Max Welling. 2017. Semi-supervised classification with graph convolutional networks. In *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*. OpenReview.net.
- Guohao Li, Matthias Müller, Bernard Ghanem, and Vladlen Koltun. 2021. Training graph neural networks with 1000 layers. In *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 6437–6449. PMLR.
- Yichuan Li, Kaize Ding, and Kyumin Lee. 2023a. GRENADE: graph-centric language model for self-supervised representation learning on text-attributed graphs. In *Findings of the Association for Computational Linguistics: EMNLP 2023, Singapore, December 6-10, 2023*, pages 2745–2757. Association for Computational Linguistics.
- Yuhan Li, Zhixun Li, Peisong Wang, Jia Li, Xiangguo Sun, Hong Cheng, and Jeffrey Xu Yu. 2023b. A survey of graph meets large language model: Progress and future directions. *CoRR*, abs/2311.12399.
- Hao Liu, Jiarui Feng, Lecheng Kong, Ningyue Liang, Dacheng Tao, Yixin Chen, and Muhan Zhang. 2024a. One for all: Towards training one graph model for all classification tasks. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net.
- Jiawei Liu, Cheng Yang, Zhiyuan Lu, Junze Chen, Yibo Li, Mengmei Zhang, Ting Bai, Yuan Fang, Lichao Sun, Philip S. Yu, and Chuan Shi. 2023. Towards graph foundation models: A survey and beyond. *CoRR*, abs/2310.11829.
- Zheyuan Liu, Xiaoxin He, Yijun Tian, and Nitesh V. Chawla. 2024b. Can we soft prompt llms for graph learning tasks? In *Companion Proceedings of the ACM on Web Conference 2024, WWW 2024, Singapore, Singapore, May 13-17, 2024*, pages 481–484. ACM.
- Haitao Mao, Zhikai Chen, Wenzhuo Tang, Jianan Zhao, Yao Ma, Tong Zhao, Neil Shah, Mikhail Galkin, and Jiliang Tang. 2024. Position: Graph foundation models are already here. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*. OpenReview.net.
- Costas Mavromatis, Vassilis N. Ioannidis, Shen Wang, Da Zheng, Soji Adeshina, Jun Ma, Han Zhao, Christos Faloutsos, and George Karypis. 2023. Train your own GNN teacher: Graph-aware distillation on textual graphs. In *Machine Learning and Knowledge Discovery in Databases: Research Track - European Conference, ECML PKDD 2023, Turin, Italy, September 18-22, 2023, Proceedings, Part III*, volume 14171 of *Lecture Notes in Computer Science*, pages 157–173. Springer.
- Andrew Kachites McCallum, Kamal Nigam, Jason Renie, and Kristie Seymore. 2000. Automating the construction of internet portals with machine learning. *Inf. Retr.*, 3(2):127–163.
- Péter Mernyei and Catalina Cangea. 2020. Wiki-cs: A wikipedia-based benchmark for graph neural networks. *CoRR*, abs/2007.02901.

- Swaroop Mishra, Daniel Khashabi, Chitta Baral, and Hannaneh Hajishirzi. 2022. [Cross-task generalization via natural language crowdsourcing instructions](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2022, Dublin, Ireland, May 22-27, 2022, pages 3470–3487. Association for Computational Linguistics.
- Seth A. Myers, Aneesh Sharma, Pankaj Gupta, and Jimmy Lin. 2014. [Information network or social network?: the structure of the twitter follow graph](#). In *23rd International World Wide Web Conference, WWW '14, Seoul, Republic of Korea, April 7-11, 2014, Companion Volume*, pages 493–498. ACM.
- Jianmo Ni, Jiacheng Li, and Julian J. McAuley. 2019. [Justifying recommendations using distantly-labeled reviews and fine-grained aspects](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 188–197. Association for Computational Linguistics.
- Norman Di Palo, Arunkumar Byravan, Leonard Hasenclever, Markus Wulfmeier, Nicolas Heess, and Martin A. Riedmiller. 2023. [Towards A unified agent with foundation models](#). *CoRR*, abs/2307.09668.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. [Learning transferable visual models from natural language supervision](#). In *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR.
- Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. [Exploring the limits of transfer learning with a unified text-to-text transformer](#). *J. Mach. Learn. Res.*, 21:140:1–140:67.
- Nils Reimers and Iryna Gurevych. 2019. [Sentence-bert: Sentence embeddings using siamese bert-networks](#). In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 3980–3990. Association for Computational Linguistics.
- Xubin Ren, Jiabin Tang, Dawei Yin, Nitesh V. Chawla, and Chao Huang. 2024. [A survey of large language models for graphs](#). In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2024, Barcelona, Spain, August 25-29, 2024*, pages 6616–6626. ACM.
- Prithviraj Sen, Galileo Namata, Mustafa Bilgic, Lise Getoor, Brian Gallagher, and Tina Eliassi-Rad. 2008. [Collective classification in network data](#). *AI Mag.*, 29(3):93–106.
- Dong Shu, Tianle Chen, Mingyu Jin, Chong Zhang, Mengnan Du, and Yongfeng Zhang. 2024. [Knowledge graph large language model \(kg-llm\) for link prediction](#). *arXiv preprint arXiv:2403.07311*.
- Juntao Tan, Shuyuan Xu, Wenyue Hua, Yingqiang Ge, Zelong Li, and Yongfeng Zhang. 2024. [Idgenrec: Llm-recsys alignment with textual ID learning](#). In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, pages 355–364. ACM.
- Jiabin Tang, Yuhao Yang, Wei Wei, Lei Shi, Lixin Su, Suqi Cheng, Dawei Yin, and Chao Huang. 2024. [Graphgpt: Graph instruction tuning for large language models](#). In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR 2024, Washington DC, USA, July 14-18, 2024*, pages 491–500. ACM.
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurélien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. 2023. [Llama: Open and efficient foundation language models](#). *CoRR*, abs/2302.13971.
- Petar Velickovic, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Liò, and Yoshua Bengio. 2017. [Graph attention networks](#). *CoRR*, abs/1710.10903.
- Petar Velickovic, William Fedus, William L. Hamilton, Pietro Liò, Yoshua Bengio, and R. Devon Hjelm. 2019. [Deep graph infomax](#). In *7th International Conference on Learning Representations, ICLR 2019, New Orleans, LA, USA, May 6-9, 2019*. OpenReview.net.
- Heng Wang, Shangbin Feng, Tianxing He, Zhaoxuan Tan, Xiaochuang Han, and Yulia Tsvetkov. 2023. [Can language models solve graph problems in natural language?](#) In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Jianing Wang, Junda Wu, Yupeng Hou, Yao Liu, Ming Gao, and Julian J. McAuley. 2024a. [Instructgraph: Boosting large language models via graph-centric instruction tuning and preference alignment](#). In *Findings of the Association for Computational Linguistics, ACL 2024, Bangkok, Thailand and virtual meeting, August 11-16, 2024*, pages 13492–13510. Association for Computational Linguistics.
- Taowen Wang, Zheng Fang, Haochen Xue, Chong Zhang, Mingyu Jin, Wujiang Xu, Dong Shu, Shanchieh Yang, Zhenting Wang, and Dongfang Liu. 2024b. [Large vision-language model security: A](#)

- survey. In *Frontiers in Cyber Security*, pages 3–22, Singapore. Springer Nature Singapore.
- Zehong Wang, Zheyuan Zhang, Nitesh V. Chawla, Chuxu Zhang, and Yanfang Ye. 2024c. [GFT: graph foundation model with transferable tree vocabulary](#). *CoRR*, abs/2411.06070.
- Jason Wei, Maarten Bosma, Vincent Y. Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2022. [Finetuned language models are zero-shot learners](#). In *The Tenth International Conference on Learning Representations, ICLR 2022, Virtual Event, April 25-29, 2022*. OpenReview.net.
- Oliver Wieder, Stefan Kohlbacher, Mélaïne Kuenemann, Arthur Garon, Pierre Ducrot, Thomas Seidel, and Thierry Langer. 2020. A compact review of molecular property prediction with graph neural networks. *Drug Discovery Today: Technologies*, 37:1–12.
- Qitian Wu, Wentao Zhao, Zenan Li, David P. Wipf, and Junchi Yan. 2022. [Nodeformer: A scalable graph structure learning transformer for node classification](#). In *Advances in Neural Information Processing Systems 35: Annual Conference on Neural Information Processing Systems 2022, NeurIPS 2022, New Orleans, LA, USA, November 28 - December 9, 2022*.
- Qitian Wu, Wentao Zhao, Chenxiao Yang, Hengrui Zhang, Fan Nie, Haitian Jiang, Yatao Bian, and Junchi Yan. 2023. [Simplifying and empowering transformers for large-graph representations](#). In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Jun Xia, Chengshuai Zhao, Bozhen Hu, Zhangyang Gao, Cheng Tan, Yue Liu, Siyuan Li, and Stan Z. Li. 2023. [Mole-bert: Rethinking pre-training graph neural networks for molecules](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Lianghao Xia, Ben Kao, and Chao Huang. 2024. [Open-graph: Towards open graph foundation models](#). *CoRR*, abs/2403.01121.
- Hao Yan, Chaozhuo Li, Ruosong Long, Chao Yan, Jianan Zhao, Wenwen Zhuang, Jun Yin, Peiyan Zhang, Weihao Han, Hao Sun, Weiwei Deng, Qi Zhang, Lichao Sun, Xing Xie, and Senzhang Wang. 2023. [A comprehensive study on text-attributed graphs: Benchmarking and rethinking](#). In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems 2023, NeurIPS 2023, New Orleans, LA, USA, December 10 - 16, 2023*.
- Ruosong Ye, Caiqi Zhang, Runhui Wang, Shuyuan Xu, and Yongfeng Zhang. 2024. [Language is all a graph needs](#). In *Findings of the Association for Computational Linguistics: EACL 2024, St. Julian’s, Malta, March 17-22, 2024*, pages 1955–1973. Association for Computational Linguistics.
- Chong Zhang, Mingyu Jin, Dong Shu, Taowen Wang, Dongfang Liu, and Xiaobo Jin. 2024a. [Target-driven attack for large language models](#). In *ECAI 2024*, pages 1752–1759. IOS Press.
- Chong Zhang, Mingyu Jin, Qinkai Yu, Chengzhi Liu, Haochen Xue, and Xiaobo Jin. 2024b. [Goal-guided generative prompt injection attack on large language models](#). In *2024 IEEE International Conference on Data Mining (ICDM)*, pages 941–946.
- Jianan Zhao, Meng Qu, Chaozhuo Li, Hao Yan, Qian Liu, Rui Li, Xing Xie, and Jian Tang. 2023a. [Learning on large-scale text-attributed graphs via variational inference](#). In *The Eleventh International Conference on Learning Representations, ICLR 2023, Kigali, Rwanda, May 1-5, 2023*. OpenReview.net.
- Jianan Zhao, Le Zhuo, Yikang Shen, Meng Qu, Kai Liu, Michael M. Bronstein, Zhaocheng Zhu, and Jian Tang. 2023b. [Graphtext: Graph reasoning in text space](#). *CoRR*, abs/2310.01089.
- Yun Zhu, Yaoke Wang, Haizhou Shi, and Siliang Tang. 2024. [Efficient tuning and inference for large language models on textual graphs](#). *CoRR*, abs/2401.15569.

APPENDIX

This appendix contains additional details for our paper, which is organized as follows:

- §A provides **Data Descriptions** used in our experiments.
- §B illustrates more details about **Baselines** employed for comparison.
Embedding-based GNNs和Prompt-based GNNs比较
- §C shows systematic comparison between **Embedding-based GNNs and Prompt-based GNNs**.
- §D analyzes **Prompt Design** in our PromptGFM framework. Prompt Design分析
- §E shows more **Implementation Details**.
- §F reports the **Hyperparameter Sensitivity** for prompt-based GNNs.
- §G presents the **Case Study** to offer deeper insights of our research.

A Data Descriptions

Dataset	Domain	#Nodes	#Edges	#Classes	Raw Text
Cora	Computer Science	2708	10858	7	paper titles and abstracts
Citeseer	Computer Science	3327	9464	6	paper titles and abstracts
Ogbn-arxiv	Computer Science	169343	2332486	40	paper titles and abstracts
WikiCS	Computer Science	11701	431726	10	wikipedia entry names and contents
Photo	E-commerce	48362	873782	12	item titles and reviews
History	E-commerce	41551	503180	12	item titles and descriptions
Pubmed	Biomedical	19717	88648	3	paper titles and abstracts

Table 8: Statistics of seven benchmarking datasets from three domains.

We utilize seven public benchmarking datasets from three domains to evaluate our framework, and the statistics of these datasets are illustrated in Table 8. For consistency, all graphs are treated as undirected. Below, we provide detailed descriptions of each dataset.

A.1 Computer Science Datasets

- **Cora (McCallum et al., 2000)**. The dataset includes a citation network consisting of 2,708 scientific publications in the field of machine learning, categorized into 7 classes based on their research topics. Nodes represent individual papers, and edges denote citation links between them, totaling 10,858 connections. Each paper is associated with its title and abstract.
- **Citeseer (Giles et al., 1998)**. This dataset introduces a citation network comprising 3,327 scientific publications, categorized into 6 classes, including Agents, Artificial Intelligence, Database, Information Retrieval, Machine Learning, and Human-Computer Interaction. Nodes correspond to documents, and edges represent citation relationships between them, amounting to 9,464 links.
- **ogbn-arxiv (Hu et al., 2020)**. The ogbn-arxiv dataset is part of the Open Graph Benchmark and consists of a directed citation graph of 169,343 arXiv papers, categorized into 40 subject areas. Nodes represent individual papers, and edges indicate citation relationships, totaling over 2.3 million connections. Each paper includes textual data from its title and abstract.
- **WikiCS (Mernyei and Cangea, 2020)**. The WikiCS dataset is a benchmark dataset derived from Wikipedia, designed for evaluating GNNs. It comprises a citation network with 11,701 nodes representing computer science articles and 431,726 edges corresponding to hyperlinks between them. The dataset features 10 classes corresponding to branches of computer science, with very high connectivity.

A.2 E-commerce Datasets

- **History (Yan et al., 2023).** The History dataset, extracted from the Amazon-Books dataset, includes items categorized under the second-level label "History". It comprises 41,511 nodes and 503,180 edges, where each node represents a book, and edges indicate frequent co-purchases or co-views between books. This dataset incorporates the title and description of each book as text attributes of the node. The classification task involves assigning books to 12 distinct categories.
- **Photo (Yan et al., 2023).** The Photo dataset, derived from the Amazon-Electronics dataset, consists of 48,362 nodes and 873,782 edges, forming a dense network that reflects user purchasing behavior. Each node represents an electronic product, while edges indicate frequent co-purchases or co-views between items. For textual attributes, user reviews are incorporated, prioritizing the most upvoted review when available; otherwise, a randomly selected review is used. The classification task involves categorizing electronic products into 12 distinct categories.

A.3 Biomedical Datasets

- **PubMed (Sen et al., 2008).** PubMed is a citation network of 19,717 scientific publications from the PubMed database pertaining to diabetes, classified into 3 classes: experimental induced diabetes, type 1 diabetes, and type 2 diabetes. Nodes are research papers, and edges signify citation links, amounting to 88,648 connections. This dataset is used for large-scale graph representation learning and evaluating algorithms in the biomedical domain.

B Baselines

We provide detailed information on the baseline models, categorized into: (1) Graph-agnostic methods, (2) GNN-based methods, (3) BERT-based methods, and (4) GNN-LLM integration methods.

B.1 Graph-agnostic methods.

- **MLP.** This method adopts a multi-layer perceptron to learn low-dimensional embeddings for each node. In our work, we randomly initialize node embeddings without textual attributes.

B.2 GNN-based methods.

- **GCN (Kipf and Welling, 2017).** This model introduces a neural network architecture that generalizes convolution operations to graph-structured data, enabling effective semi-supervised learning by aggregating feature information from a node’s local neighborhood.
- **GAT (Velićković et al., 2017).** This method incorporates attention mechanisms into graph neural networks, allowing nodes to assign different importance weights to their neighbors during feature aggregation, which enhances performance by focusing on the most relevant connections.
- **SAGE (Hamilton et al., 2017).** GraphSAGE is an inductive representation learning framework on large graphs; it generates node embeddings by sampling and aggregating features from a node’s local neighborhood, facilitating generalization to unseen nodes or graphs.
- **ReVGNN (Li et al., 2021).** This method includes a recurrent graph neural network tailored for dynamic graphs, capturing temporal dependencies by updating node representations as events occur over time, which is crucial for modeling evolving graph structures.
- **SGFormer (Wu et al., 2023).** This work introduces a transformer-based architecture designed for graph data, integrating spectral graph theory into the transformer framework. It aims to capture both local and global graph structures efficiently by incorporating spectral filters, enhancing the model’s ability to learn complex graph representations.
- **NodeFormer (Wu et al., 2022).** This framework presents a scalable graph transformer model that utilizes a randomized attention mechanism to approximate full attention on graphs. By reducing computational complexity, it enables efficient learning on large-scale graphs while preserving the expressiveness of transformer architectures.

B.3 BERT-based methods.

- **BERT (Devlin et al., 2019).** This work introduces a deep learning model that understands language context by processing text bidirectionally. It is pre-trained using masked language modeling (MLM) and next sentence prediction (NSP) to learn rich linguistic features. BERT can be fine-tuned for various NLP tasks, achieving state-of-the-art performance.
- **Sentence-BERT (Reimers and Gurevych, 2019).** This model is a modification of BERT designed for sentence embeddings, allowing efficient comparison of semantic similarity. It uses a siamese or triplet network structure to generate fixed-size vector representations, making tasks like sentence similarity, clustering, and retrieval significantly faster. Unlike BERT, which requires computationally expensive cross-encoding, Sentence-BERT enables quick and effective comparisons using cosine similarity.

B.4 GNN-LLM integration methods.

- **LLaGA (Chen et al., 2024a).** This model effectively integrates LLM capabilities to handle the complexities of graph-structured data. It transforms graph nodes into structure-aware sequences and maps them into token embedding space using a specialized projector. LLaGA excels in generalization and interpretability, performing strongly across various datasets and tasks. It also supports zero-shot learning, making it highly adaptable for unseen datasets.
- **OFA (Liu et al., 2024a).** This paper proposes a framework that handles various graph classification tasks across different domains using a single model. It introduces the nodes-of-interest (NOI) subgraph mechanism to standardize different tasks with a single task representation. Additionally, a novel graph prompting paradigm to leverage in-context learning and apply the same architecture across diverse graph classification tasks, achieving generalization across multiple domains.
- **GraphPrompter (Liu et al., 2024b).** This work introduces a novel framework designed to align graph with LLMs via soft prompts. Specifically, it adopts GNNs to capture graph structure and leverages an LLM to interpret the textual information at the node level. By prompt tuning, this approach demonstrates the potential of LLMs to effectively interpret graph structures, combining both semantic and structural insights for improved graph learning tasks.
- **ENGINE (Zhu et al., 2024).** This paper proposes a parameter- and memory-efficient fine-tuning method for textual graphs by using LLMs as encoders. It combines the LLMs and GNNs through a tunable GNN-based side structure, called G-Ladder, alongside each LLM layer, effectively reducing training costs without compromising performance.

C Embedding-based GNNs vs. Prompt-based GNNs

Table 6 illustrates the details of the graph understanding module, with the prompt-based GNN as its core component. To ensure clarity, we systematically compare traditional embedding-based GNNs with our proposed prompt-based GNN, emphasizing its advantages.

- **Input and output.** In the embedding-based GNN framework, for each node, *structure-less embeddings* are progressively refined to *structure-rich embeddings*, whereas *verbose textual sequences* are gradually converted to *concise textual sequences* in our prompt-based GNN.
- **Message passing.** The *multi-layer embedding updates* are mirrored by *multi-round LLM calls* in text space, both of which progressively refining the representations in different spaces.
- **Neighbor sampling.** The *neighbor sampling* operation used to reduce computational load in traditional embedding-based GNNs is analogous to the *selected one-hop neighbor descriptions* employed to address prompt length limitations in prompt-based GNNs.
- **Aggregation-update mechanism.** Embedding-based GNNs use *predefined operators* (e.g., mean aggregator, weighted aggregator, or LSTM aggregator) to achieve message passing in the embedding space, while prompt-based GNNs use *straightforward prompts* to guide LLMs in executing the process more flexibly without predefined rules.
- **Optimization.** In prompt-based GNNs, we use heuristic prompts at each layer to reflect the key idea of contrastive loss. These *cumulative layer-by-layer prompts* are comparable to the layer-wise loss combination, formally as *mean pooling*, commonly seen in embedding-based GNNs.

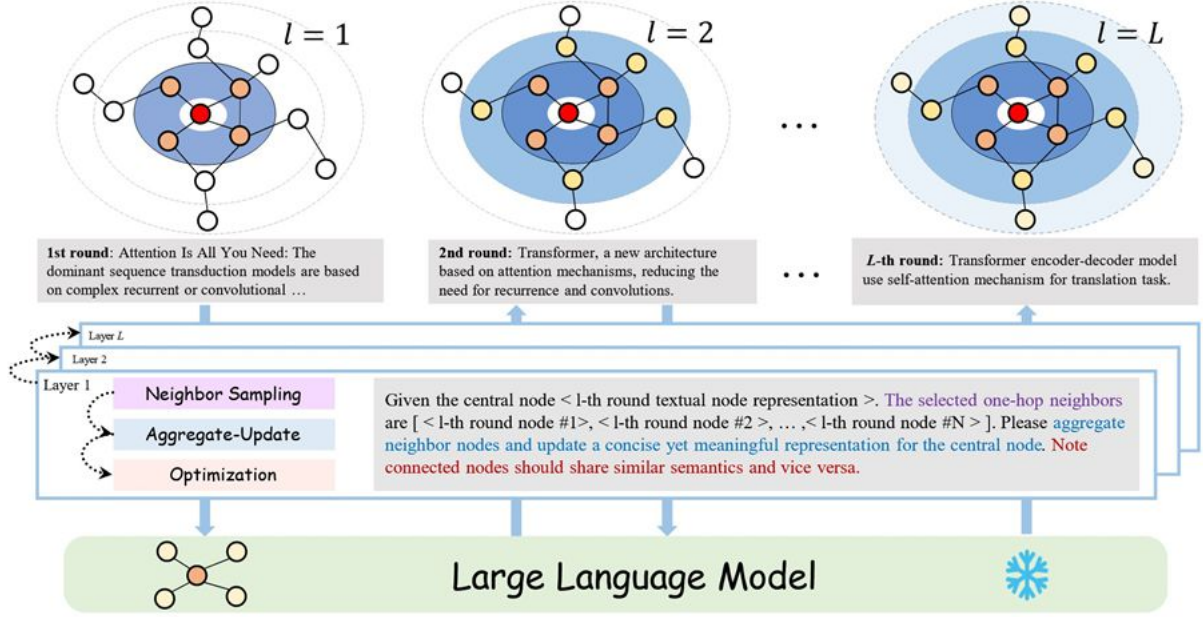


Figure 6: Graph understanding module via prompt-based GNNs. We prompt LLMs to achieve fine-grained reproduction of traditional GNN workflow, refining verbose textual representations into concise yet meaningful ones. In the prompt, neighbor sampling (see Equation 3) is highlighted in purple, the aggregation-update mechanism (see Equation 4) in blue, and the optimization in red.

In conclusion, our approach enables LLMs to replicate the finest GNN workflow within the text space, fostering seamless graph-text alignment. By bridging structural and semantic information through language-based representations, this framework introduces a novel paradigm for GNN-LLM integration. It not only enhances the interpretability and transferability of graph learning but also unlocks new possibilities for leveraging LLMs in graph-related tasks without relying on numerical embeddings.

D Prompt Design

In this section, we provide the templates of prompts in our PromptGFM framework, including prompt-based GNNs and task-oriented prompts.

Prompt for node initialization.

The title of the paper is <the title of the paper>, the abstract of the paper is <the abstract of the paper>. Please summarize the paper.

/** This prompt varies depending on the dataset. The instance above is designed for citation datasets. **/

Prompt for each layer of GNN replication.

Given the central node <l-th round textual representation of the central node>. The selected one-hop neighbors are [<l-th round of node #1>, <l-th round node #2>, ... ,<l-th round node #N>]. Please aggregate neighbor nodes and update a concise yet meaningful representation for the central node. Note connected nodes should share similar semantics and vice versa.

Prompt for node classification.

<the language-based ID of the central node> has 1-hop connections with [..., <language-based IDs of its 1-hop neighbors>, ...], and it also has 2-hop connections with [..., <language-based IDs of its 2-hop neighbors>, ...]. Which category should <the language-based ID of the central node> be classified as ?

Prompt for discriminative link prediction.

<the language-based ID of the central node> has 1-hop connections with [..., <language-based IDs of its 1-hop neighbors>, ...], and it also has 2-hop connections with [..., <language-based IDs of its 2-hop neighbors>, ...]. Among <the language-based ID of the central node> and <the language-based ID of its negative sampling node>, which node will be connected to <the language-based ID of the central node>?

Prompt for generative link prediction.

<the language-based ID of the central node> has 1-hop connections with [..., <language-based IDs of its 1-hop neighbors>, ...], and it also has 2-hop connections with [..., <language-based IDs of its 2-hop neighbors>, ...]. Which node should be connected to <the language-based ID of the central node>?

E Implementation Details

We provide further information for reproduction. In the graph understanding module, we selected the number of layers for the prompt-based GNN from $\{1, 2, 3, 4\}$. We randomly sampled 30% of the first-order neighbors during neighborhood sampling, capping the maximum number of sampled nodes at 20 to reduce computational cost and prevent overfitting. In the graph inference module, we fine-tuned the LLM with a learning rate of $3e-4$ and a batch size of 4. To mitigate potential biases introduced by task-specific prompts, we designed a prompt pool for each task requirement and randomly selected prompts during instruction construction to enhance robustness. We employ 10-fold cross-validation and report average results across all folds. We employed a standard early-stopping strategy during training: if the performance metric on the validation set did not improve over a fixed number of consecutive epochs (determined based on the dataset), we halted training to prevent overfitting. For other hyperparameters of the compared methods, we referred to the original papers and carefully tuned them to suit each dataset. In terms of the TAG setting, we utilize the textual features for initialization in all embedding-based models.

F Hyperparameter Sensitivity

We explore the impact of the number of layers in our prompt-based GNN. As shown in Figure 7, we can observe that PromptGFM progressively improves as the layer increases due to its ability to capture broader context and higher-order relationships over the graph. However, after a certain point, further stacking layers results in diminishing returns or even performance degradation due to over-smoothing, where

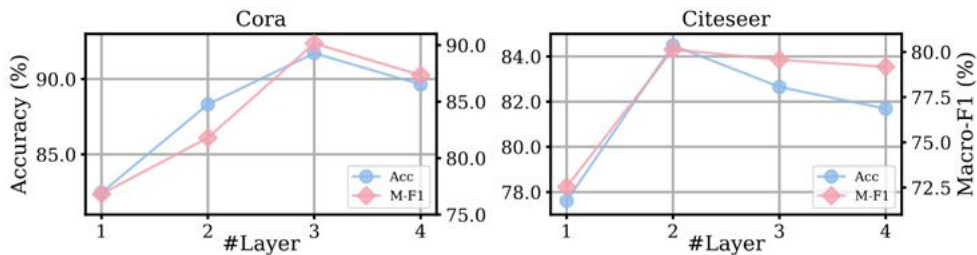


Figure 7: Impact of varying prompt-based GNN layers on node classification performance.

node representations become indistinguishable within their local structures. This trend is consistent with traditional GNNs. Optimal performance is achieved with 3-layer GNN for Cora, while Citeseer reaches its best results with 2 layers. This analysis suggests that textual representations can be propagated over the graph similarly to numerical embeddings, effectively capturing semantic and structural information simultaneously.

G Case Study

G.1 Textual Representations in Prompt-based GNNs

In this part, we select two representative cases in citation networks and demonstrate their layer-by-layer refinement of our prompt-based GNN. Specifically, we provide the textual representations at each layer, including Round 0 as initial features. More importantly, we also collect its one-hop nodes and highlighted the source and relevant information below. From our empirical studies, we have the following observations. Overall, it is evident that the verbose textual representations are progressively refined to concise textual presentations. Meanwhile, the core semantics become increasingly clear throughout the process, until a short sequence composed of several natural language tokens at the last round. Furthermore, we notice that we effectively incorporate the key ideas of some neighboring nodes, as reflected in the refined textual representations after each aggregation-update operation. More specifically, the neighboring information is naturally fused to the central node within the text space, boosting their connections and improving the expressiveness of the central node. This study further illustrates the explicit message passing through our proposed prompt-based GNN in the text space, which captures the graph semantics and structures simultaneously.

G.1.1 Case A

Raw Text Attributes.

Paper Title: Evaluating Neural Network Predictors by Bootstrapping. Abstract: We present a new method, inspired by the bootstrap, whose goal it is to determine the quality and reliability of a neural network predictor. Our method leads to more robust forecasting along with a large amount of statistical information on forecast performance that we exploit. We exhibit the method in the context of multi-variate time series prediction on financial data from the New York Stock Exchange. It turns out that the variation due to different resamplings (i.e., splits between training, cross-validation, and test sets) is significantly larger than the variation due to different network conditions (such as architecture and initial weights). Furthermore, this method allows us to forecast a probability distribution, as opposed to the traditional case of just a single value at each time step. We demonstrate this on a strictly held-out test set that includes the 1987 stock market crash. We also compare the performance of the class of neural networks to identically bootstrapped linear models.

Round 0: Textual Feature Initialization.

This method evaluates neural network predictors using a bootstrap-inspired approach, improving robustness in NYSE financial data forecasting. It predicts probability distributions rather than single values, surpassing resampling variations. Compared to bootstrapped linear models, it enhances accuracy, as demonstrated during the 1987 stock market crash.

Round 1: 1st Layer Representations.

This bootstrap method improves neural network evaluations in NYSE forecasting, **comparable to traditional error estimation methods**. It clears and simplifies data structures, enhancing accuracy and predicting probability distribution.

/***"comparable to traditional error estimation methods" comes from the paper: *A comparison of some error estimates for neural network models*. ***/

/***"It clears and simplifies data structures" comes from the paper: *The Observer-Observation Dilemma in Neuro-Forecasting: Reliable Models From Unreliable Data Through CLEARNING*. ***/

Round 2: 2nd Layer Representations.

This bootstrap-inspired method refines neural network evaluations for NYSE forecasting, highlighting **local error bars from nonlinear regression analysis**, improving accuracy with data simplification techniques.

/***"local error bars from nonlinear regression analysis" comes from the paper: *Learning Local Error Bars for Nonlinear Regression*. ***/

Round 3: 3rd Layer Representations.

Bootstrap method refines neural forecasts, integrates error bars, enhances accuracy.

G.1.2 Case B

Raw Text Attributes.

Paper Title: Insulin dependent diabetes in Asians. Abstract: Type 1 diabetes is said to be extremely rare in children in India, where diabetes treated with insulin may be due to chronic pancreatic disease or malnutrition. To see whether typical type 1 diabetes occurred in Asian children in the United Kingdom, all known Asian children with diabetes in industrial West Yorkshire were ascertained. A total of 17 such children were studied; of these, seven were from three multiplex families and two fathers from these families had diabetes. All children were ketosis prone and developed diabetes while resident in the UK. There were significant increases in HLA-B8 and HLA-DR3 and increases in HLA-DR4 and HLA-DR3/DR4, while HLA-B15 was absent. Islet cell antibodies, either IgG or complement fixing, were present in four of 18 subjects tested, all of whom had disease of short duration. The prevalence of type 1 diabetes in Asian children aged 15 years or less in West Yorkshire was 36/100,000, assuming complete ascertainment. It is concluded that typical type 1 diabetes may occur in Asian children and this condition may be more common in families who have migrated to the UK.

Round 0: Textual Feature Initialization.

The study examined type 1 diabetes prevalence in 17 Asian children in West Yorkshire, UK, focusing on family history. Seven were from multiplex families, and all were ketosis-prone. Significant increases in HLA-B8, HLA-DR3, and HLA-DR4 were found, with no HLA-B15. Islet cell antibodies were present in four of 18 subjects. The diabetes prevalence was 36/100,000, suggesting it may be more common in migrant families.

Round 1: 1st Layer Representations.

The study examined type 1 diabetes in Asian children in West Yorkshire, UK. **Environmental factors and migration may raise incidence**, unlike the **low incidence of insulin-dependent diabetes in Karachi**. Increases in HLA-B8, HLA-DR3, and HLA-DR4 were found.

/*****"Environmental factors and migration may raise incidence"** comes from the paper: *Evidence for an environmental effect in the aetiology of insulin dependent diabetes in a trans migratory population*. ***/

/*****"low incidence of insulin-dependent diabetes in Karachi"** comes from the paper: *Incidence of insulin dependent diabetes mellitus in Karachi, Pakistan*. ***/

Round 2: 2nd Layer Representations.

The study examined type 1 diabetes in Asian children in West Yorkshire, UK. **Migration may raise incidence**, unlike the low incidence in Karachi. Increases in HLA markers were found, similar to **North and South Indian diabetics**.

/*****"Migration may raise incidence"** comes from the paper: *HLA-DR antigen frequencies in a North Indian type I diabetic population*. ***/

/*****"North and South Indian diabetics"** comes from two papers: *HLA-DR antigen frequencies in a North Indian type I diabetic population* and *HLA, complement C2, C4, properdin factor B and glyoxalase types in South Indian diabetics*. ***/

Round 3: 3rd Layer Representations.

Type 1, diabetes. Migration and HLA markers linked to increased diabetes incidence.

G.2 Language-based IDs vs. Key Words

To provide further insights, we leverage external information to validate the superiority of PromptGFM in capturing the core semantics of nodes. Specifically, we extract key words of papers from citation datasets (Cora, Citeseer, Obgn-arxiv, and PubMed) and compare them with their language-based IDs in our universal graph vocabulary.

Table 9 summarizes the key words and language-based IDs of selected papers, along with their titles and URLs for reference. Overall, it is evident that there are strong semantic relevance between the language-based IDs and keywords. For example, regarding the paper titled *Distributed Protocols at the Rescue for Trustworthy Online Voting*, the key words have appeared within its language-based ID, suggesting that PromptGFM has effectively captured its core semantics through our prompt-based GNN. In addition, in *Committees providing EJR can be computed efficiently*, where the title is less indicative of the content, the language-based ID still aligns perfectly with the corresponding key words, such as *efficient computation* and *rules*. This finding demonstrates that PromptGFM not only effectively captures the core idea without relying on the title, but also filters relevant semantics from neighboring nodes to enhance its own representations. Overall, our language-based IDs accurately capture and extend the semantics of the nodes, making them well-suited to form a universal graph vocabulary.

Table 9: The comparison between language-based IDs generated by prompt-based GNNs as part of the graph vocabulary and keywords extracted from the original papers in citation datasets. To prevent data leakage, we exclude these keywords from model training. The observed similarity confirms that our prompt-based GNN effectively captures the essential semantics of nodes. Furthermore, our language-based IDs encode richer information than keywords by preserving high-order structural signals.

Paper	Language-based ID	Key Words	URL
Modular Verification of Interrupt-Driven Software	Modular verification of interrupt-driven software using abstract interpretation	Software, Abstract Interpretation, Feasibility Verification	arXiv:1709.10078
Parsimonious Data: How a single Facebook like predicts voting behaviour in multiparty systems	Predicting voting behavior using Facebook likes in multiparty systems	Facebook Likes, Voter Intention, Machine Learning, Multiparty System	arXiv:1704.01143
A Fast Noniterative Algorithm for Compressive Sensing Using Binary Measurement Matrices	Fast noniterative algorithm for compressive sensing with binary matrices	Compressive Sensing, Deterministic Methods	arXiv:1708.03608
Optimization of Battery Energy Storage to Improve Power System Oscillation Damping	Battery storage optimization improves power system oscillation damping	Battery Energy Storage System, Oscillation Damping	arXiv:1811.10213
Neural Variational Hybrid Collaborative Filtering	Neural Variational Hybrid Collaborative Filtering improves recommendation performance	Collaborative Filtering, VAE, Recommendation System	arXiv:1810.05376v6
Interpretable Neural Networks for Predicting Mortality Risk using Multi-modal Electronic Health Records	Predicting mortality risk using interpretable multi-modal neural network	Mortality Risk Prediction, Clinical Data	arXiv:1901.08125
A New Approach to Distributed Hypothesis Testing and Non-Bayesian Learning: Improved Learning Rate and Byzantine-Resilience	Distributed hypothesis testing with Byzantine resilience using Bayesian update	Bayesian Learning, Byzantine Resilience	arXiv:1907.03588
Accurate and Efficient Hyperbolic Tangent Activation Function on FPGA using the DCT Interpolation Filter	Efficient hyperbolic tangent activation function using DCTIF	Hyperbolic Tangent, Activation Function	arXiv:1609.07750
Distributed Protocols at the Rescue for Trustworthy Online Voting	Trustworthy online voting with distributed blockchain protocols	Distributed Voting, Distributed Protocols	arXiv:1705.04480
Committees providing EJR can be computed efficiently	Efficient computation of approval-based multi-winner voting rules	Approval-Based Voting, Multi-Winner Elections	arXiv:1704.00356
Creatism: A deep-learning photographer capable of creating professional work	Creatism: deep learning system for artistic photography creation	Creatism, Evaluation of Photographic Quality, Deep Learning	arXiv:1707.03491
Relation of familial patterns of coronary heart disease, stroke, and diabetes to subclinical atherosclerosis: the multi-ethnic study of atherosclerosis	Family history beyond early-onset heart disease impacts atherosclerosis	Family History, Coronary Heart Disease, Stroke	doi.org/10.1097/GIM.0b013e31818e639b
Glycemic index, glycemic load, and risk of type 2 diabetes	Benefits of low-GI diet in type 2 diabetes	Diabetes, Prevention	doi.org/10.1093/ajcn/76/1.274S
Decreased insulin responsiveness of glucose uptake in cultured human skeletal muscle cells from insulin-resistant nondiabetic relatives of type 2 diabetic families	Inherited defects contribute to insulin resistance in diabetes	Insulin Resistance, Inherited Factors	doi.org/10.2337/diabetes.49.7.1169
Quantitative histopathological studies of the extramural coronary arteries from Type 2 (non-insulin-dependent) diabetic patients	Histopathological study of coronary arteries in diabetic patients	Histopathology, Diabetes Mellitus	doi.org/10.1007/BF00274798
Metabolic control and diet in Finnish diabetic adolescents	Factors influencing metabolic control in diabetic adolescents	Diabetes Mellitus, Adolescent	doi.org/10.1111/j.1651-2227.1992.tb12212.x